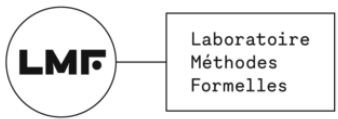
Variants of Higher-Dimensional Automata

Hugo Bazille[1] Jérémy Dubut[2] Uli Fahrenberg[3] Krzysztof Ziemiański[4]

EPITA Research Lab (LRE), Rennes/Paris, France
LIX, École polytechnique, Palaiseau, France
LMF, Université Paris-Saclay, France
Institute of Mathematics, University of Warsaw, Poland

ICTAC 2026

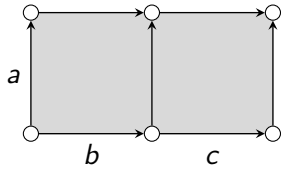


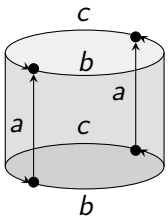
Higher-Dimensional Automata?

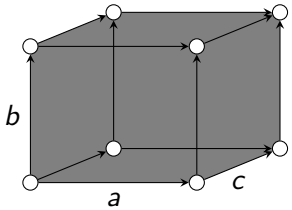
- Languages of Higher-Dimensional Automata [MSCS 2021]
- Posets with Interfaces as a Model for Concurrency [I&C 2022]
- Decision and Closure Properties for Higher-Dimensional Automata [ICTAC 2023]
- Bisimulations and Logics for Higher-Dimensional Automata [ICTAC 2024]
- Petri Nets and Higher-Dimensional Automata [Petri Nets 2025]
- ...

Higher-Dimensional Automata?

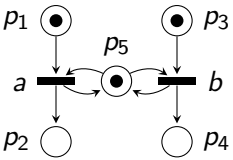
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- ...
- Rob van Glabbeek 2006: *On the expressiveness of higher-dimensional automata*



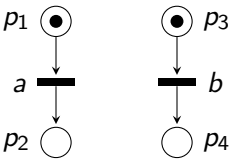




Motivation: Semantics of Petri Nets

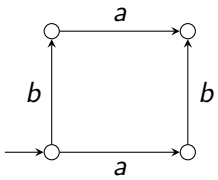


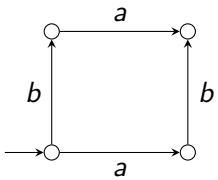
mutual exclusion



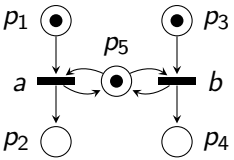
concurrency

interleaving semantics

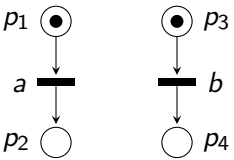




Motivation: Semantics of Petri Nets

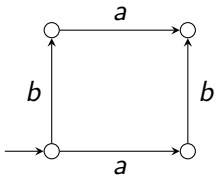


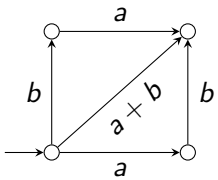
mutual exclusion



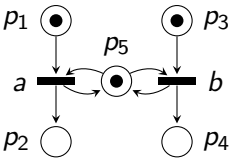
concurrency

concurrent step semantics

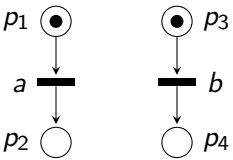




Motivation: Semantics of Petri Nets

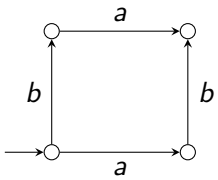


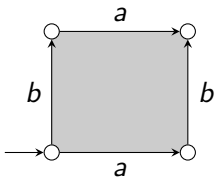
mutual exclusion



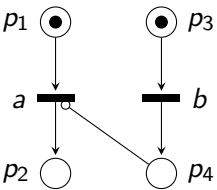
concurrency

HDA semantics



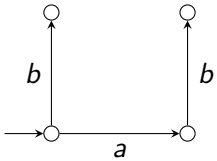


Petri Nets with Inhibitor Arcs

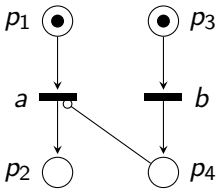


when there is a token in p_4,
then a cannot fire

a-posteriori semantics

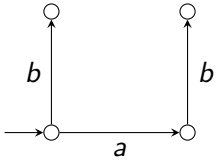


Petri Nets with Inhibitor Arcs

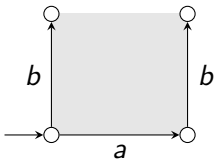


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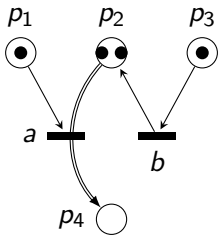


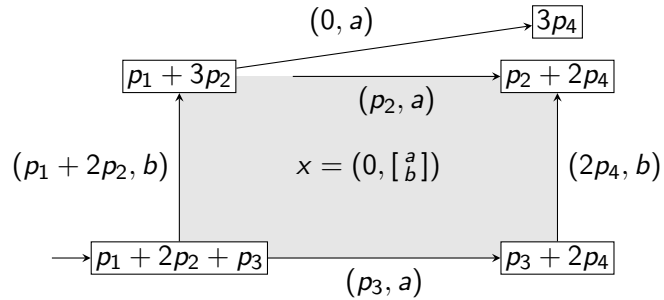
a-priori semantics



upper face of square is missing:
a partial HDA

Petri Nets with Transfer Arcs





all faces present, but face relation broken:
also a partial HDA

Concurrency Concurrency Concurrency

- Concurrency interacts in funny ways with reachability in Petri net extensions
- For Petri nets, the literature knows concurrent semantics only for read arcs and inhibitor arcs
- Not for transfer arcs or other extensions
- For real-time, even more interesting (and very little literature)
- (Partial) HDA semantics provides tools for reasoning about such interactions

1. Motivation
2. Partial Higher-Dimensional Automata
3. Other Models; Results
4. Conclusion

Higher-Dimensional Automata

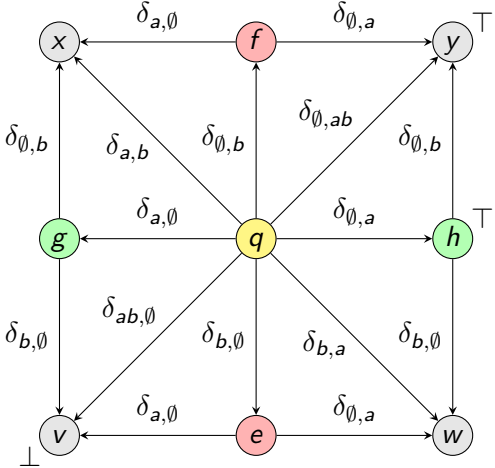
A conclist is a finite, totally ordered, Σ-labeled set. (a list of labeled events)

A precubical set X consists of:

- A set of cells X (cubes)
- Every cell $x \in X$ has a conclist $\mathrm{ev}(x)$ (list of events active in x)
- We write $X[U] = \{x \in X \mid \mathrm{ev}(x) = U\}$ for a conclist U (cells of type U)
- For every conclist U and $A, B \subseteq U$ with $A \cap B = \emptyset$ there is a face map $\delta_{A,B} : X[U] \to X[U \setminus (A \cup B)]$ (terminating events B; "unstarting" events A)
- Precubical identity: $\delta_{C,D}\delta_{A,B} = \delta_{A \cup C, B \cup D}$

A higher dimensional automaton (HDA) is a precubical set X with initial cells $\bot \subseteq X$ and accepting cells $\top \subseteq X$ (not necessarily vertices)

Example



$X[\emptyset] = \{v, w, x, y\}$

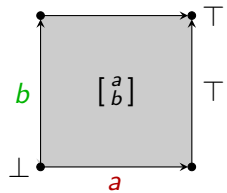
$X[a] = \{e, f\}$

$X[b] = \{g, h\}$

$X[[\begin{smallmatrix} a \\ b \end{smallmatrix}]] = \{q\}$

$\bot_X = \{v\}$

$\top_X = \{h, y\}$



Higher-Dimensional Automata & Concurrency Theory

HDAs as a model for concurrency:

- points: states
- edges: transitions
- squares, cubes etc.: independency relations / concurrently executing events
- two-dimensional automata $\cong$ asynchronous transition systems [Bednarczyk'88]
- Introduced in [van Glabbeek'89]
- [van Glabbeek'06]: translations from Petri nets and other models
  - *up to history-preserving bisimulation, HDAs generalize most known models for concurrency*

Partial Higher-Dimensional Automata

A conclist is a finite, totally ordered, Σ-labeled set. (a list of labeled events)

A partial precubical set X consists of:

- A set of cells X (cubes)
- Every cell $x \in X$ has a conclist $\mathrm{ev}(x)$ (list of events active in x)
- We write $X[U] = \{x \in X \mid \mathrm{ev}(x) = U\}$ for a conclist U (cells of type U)
- For every conclist U and $A, B \subseteq U$ with $A \cap B = \emptyset$ there is a partial face map $\delta_{A,B} : X[U] \rightharpoonup X[U \setminus (A \cup B)]$ (terminating events B; "unstarting" events A)
- Precubical inclusion: $\delta_{C,D}\delta_{A,B} \subseteq \delta_{A \cup C, B \cup D}$

A partial higher dimensional automaton (pHDA) is a partial precubical set X with initial cells $\bot \subseteq X$ and accepting cells $\top \subseteq X$ (not necessarily vertices)

Partial Higher-Dimensional Automata

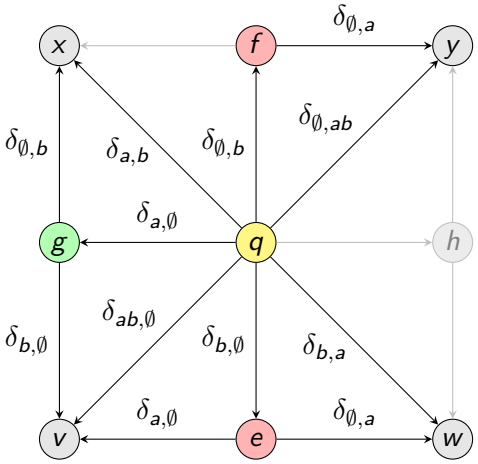
A conclist is a finite, totally ordered, Σ-labeled set. (a list of labeled events)

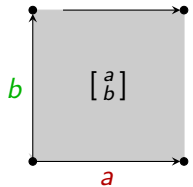
A partial precubical set X consists of:

- A set of cells X (cubes)
- Every cell $x \in X$ has a conclist $\mathrm{ev}(x)$ (list of events active in x)
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- Precubical inclusion: $\delta_{C,D}\delta_{A,B} \subseteq \delta_{A \cup C, B \cup D}$ (if defined, then equal)

A partial higher dimensional automaton (pHDA) is a partial precubical set X with initial cells $\bot \subseteq X$ and accepting cells $\top \subseteq X$ (not necessarily vertices)

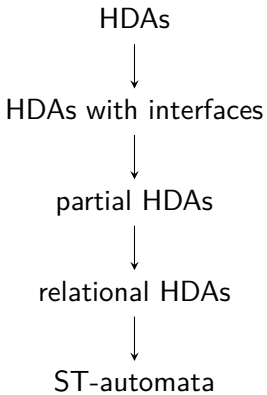
Example



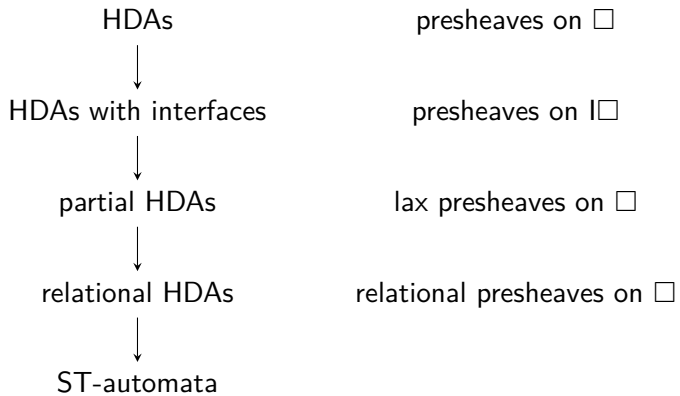


- right edge is missing: b must terminate before a
- top edge has no connection to top-left vertex: if b terminates before a has started, deadlock

The Zoo of Models



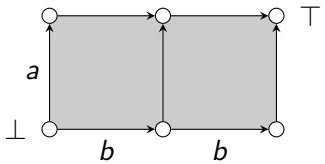
The Zoo of Models

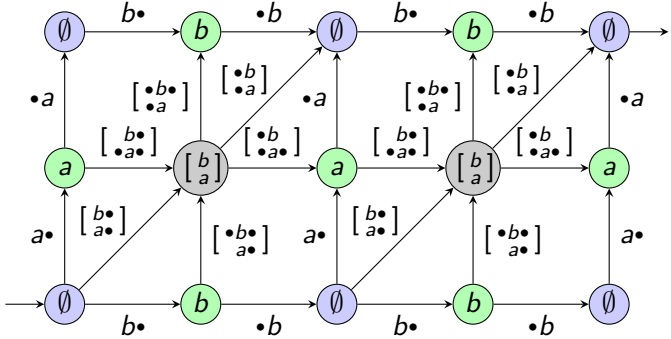


ST-Automata

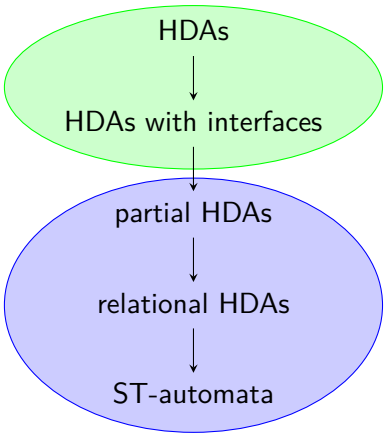
Automata on graph alphabet consisting of starters and terminators

- HDA and ST-automaton accepting $a \parallel bb$:





Languages

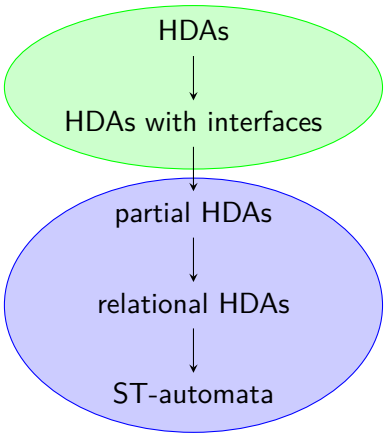


languages are rational sets of ipomsets
closed under order refinement

languages are rational sets of ipomsets

- Notably, every ST-automaton may be converted to a pHDA with the same language

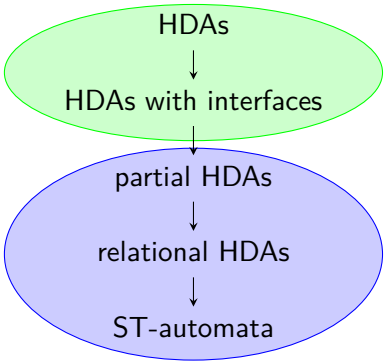
Kleene Theorem



Kleene theorem with a difficult topological proof

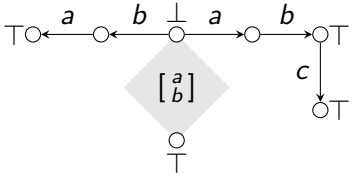
Kleene theorem with an easy standard proof using ST-automata

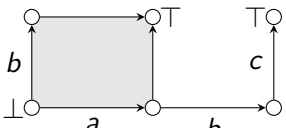
Determinisation

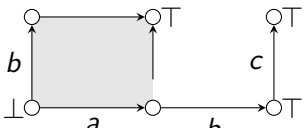


not determinizable

determinizable







Conclusion



- Higher-dimensional automata (HDAs): a nice geometric model for concurrency
- Partial HDAs: a nice algebraic model for concurrency
  - same for relational HDAs and ST-automata
- Here we've worked out their relationship, using presheaves
- Useful for passing between geometric and algebraic reasoning
- Lots of things going on in HDA-space: new national French project CHAAT
  - two PhD students almost done: Safa Zouari, Enzo Erlich
  - four new PhD students: Jakub Hajdas, Clément Borne, Safia Tehami, Lory Soumphonphakdy
  - branching-time and linear-time logics for HDAs
  - HDAs over infinite ipomsets; games on HDAs; active learning of HDAs; HDtimedA; ...